MambaHSI: Spatial–Spectral Mamba for Hyperspectral Image Classification

Yapeng Li, *Graduate Student Member, IEEE*, Yong Luo, *Member, IEEE*, Lefei Zhang, *Senior Member, IEEE*, Zengmao Wang, *Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Transformer has been extensively explored for hyperspectral image (HSI) classification. However, transformer poses challenges in terms of speed and memory usage because of its quadratic computational complexity. Recently, the Mamba model has emerged as a promising approach, which has strong long-distance modeling capabilities while maintaining a linear computational complexity. However, representing the HSI is challenging for the Mamba due to the requirement for an integrated spatial and spectral understanding. To remedy these drawbacks, we propose a novel HSI classification model based on a Mamba model, named MambaHSI, which can simultaneously model long-range interaction of the whole image and integrate spatial and spectral information in an adaptive manner. Specifically, we design a spatial Mamba block (SpaMB) to model the long-range interaction of the whole image at the pixel-level. Then, we propose a spectral Mamba block (SpeMB) to split the spectral vector into multiple groups, mine the relations across different spectral groups, and extract spectral features. Finally, we propose a spatial–spectral fusion module (SSFM) to adaptively integrate spatial and spectral features of a HSI. To our best knowledge, this is the first image-level HSI classification model based on the Mamba. We conduct extensive experiments on four diverse HSI datasets. The results demonstrate the effectiveness and superiority of the proposed model for HSI classification. This reveals the great potential of Mamba to be the next-generation backbone for HSI models. Codes are available at https://github.com/li-yapeng/MambaHSI.**

***Index Terms*— Hyperspectral image (HSI) classification, Mamba, state-space models (SSMs), transformer.**

I. Introduction

With the development of Earth observation technology, hyperspectral image (HSI) can be widely acquired [1]. Different from the traditional vision system that capture images only with the RGB channels, HSIs can cover a much larger spectral range from the visible spectrum, near-infrared, mid-infrared to far-infrared with dozens or even hundreds of continuous wavebands [2], [3], [4]. Benefiting from the abundant and detailed spectral information, HSI provides an opportunity to identify objects which are difficult to distinguish in natural RGB images [5]. This makes more accurate observations of the Earth possible [3]. Owing to the advantages mentioned above, HSIs are widely applied to diverse applications [6], such as urban mapping, resource exploring, environmental monitoring, and so on. As the fundamental task of these applications, HSI classification is to predict the class label of each pixel in the image [7].

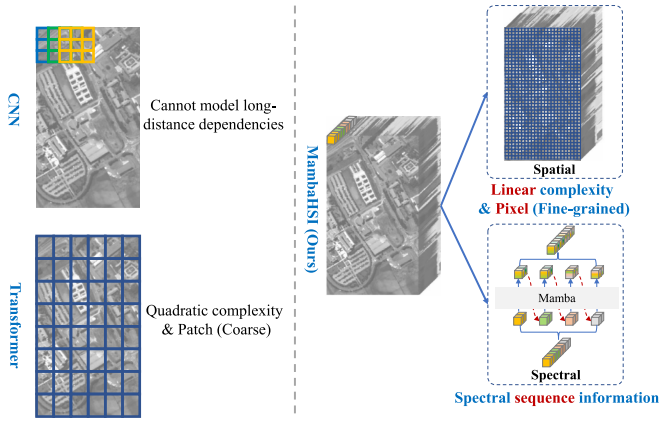


Fig. 1. Motivation illustration. The local characteristics of CNNs and the quadratic complexity of Transformers limit their ability to achieve fine-grained global modeling. In contrast, the proposed MambaHSI model can achieve pixel-level fine-grained spatial feature modeling with linear complexity. By incorporating spectral sequence information, MambaHSI enhances the extraction of spectral features.

Benefiting from the development of deep learning technology, HSI classification has witnessed a rapid advancement in the past few years [6], [8]. HSI classification methods can be generally divided into traditional machine learning (ML-based) [9], [10] and deep learning (DL-based) [11], [12]. DL-based can be roughly fall into graph convolutional network (GCN-based) [11], [13], convolutional neural network (CNN-based) [14], [15], and Transformer-based [12], [16] methods. However, as shown in Fig. 1, the mainstream HSI classification methods based on CNN and Transformer have inherent limitations. CNN-based models are constrained by their local receptive field, which hinders their ability to model long-range dependencies [17]. Although Transformer-based models show superior performance for global modeling, the self-attention mechanism of the transformer layer demands quadratic complexity in terms of the sequence length [18].

Manuscript received 24 April 2024; revised 5 July 2024; accepted 10 July 2024. Date of publication 19 July 2024; date of current version 1 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62225113 and Grant U23A20318 and in part by the Innovative Research Group Project of Hubei Province under Grant 2024AFA017. *(Corresponding authors: Zengmao Wang; Bo Du.)*

The authors are with the National Engineering Research Center for Multimedia Software, School of Computer Science, Institute of Artificial Intelligence, and Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan 430072, China (e-mail: yapengli@whu.edu.cn; yluo180@gmail.com; zhanglefei@whu.edu.cn; wangzengmao@whu.edu.cn; dubo@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3430985

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

This results in a high computational burden [19] and impedes the extraction of fine-grained spatial features at the pixel level. The drawbacks of these models compel us to develop a novel framework for HSI classification, capable of modeling strong long-range dependencies and maintaining linear computational complexity.

Recently, state-space models (SSMs) [20], [21], in particular structured state-space sequence models (S4) [22], have emerged as an efficient and effective layer to construct deep networks, and have obtained superior performance in continuous long-sequence data analysis [22], [23]. Mamba [24] further improved S4 with a selective mechanism, which enables the model to select relevant information in an input-dependent manner. Besides, Mamba achieves higher computational efficiency than Transformer by combining with hardware-aware implementation [24]. Benefiting from the strong long-range modeling capabilities while maintaining linear computational complexity, the Mamba has received substantial research across many fields, such as language understanding [22], [24], medical image analysis [17], [25], computer vision [19], [26], and so on [27], [28], [29]. However, the aforementioned methods cannot be applied to HSIs because these methods ignore the spectral continuity of HSIs.

In this article, we propose a novel HSI classification framework based on Mamba, termed as MambaHSI, which not only models the long-range dependencies but also comprehensively utilizes the spatial and spectral information (see Fig. 1). Specifically, MambaHSI is a pure-SSM-based model, which takes the whole image as inputs and can be trained in an end-to-end manner. The pure-SSM-based structure enables model to capture the long-range dependencies while maintaining the linear computational complexity. The end-to-end training method enables the model to be optimized as a whole to obtain better parameters. Then, considering the spatial and spectral characteristics of HSIs, we design a spatial and spectral Mamba block (SpeMB) to extract spatial and spectral information, respectively. Benefiting from the strong long-range modeling ability, the spatial and spectral Mamba block can capture more discriminative spatial and spectral features for classification. To comprehensively exploit the spatial and spectral information, we propose a spatial–spectral fusion module (SSFM), which can adaptively fuse the spatial and spectral information based on the importance of spatial and spectral. The idea of residual learning has also been introduced to help with model learning.

To Summarize, the main contributions of this article are as follows.

1) To our best knowledge, the proposed MambaHSI is the first image-level HSI classification model based on SSM, which can simultaneously model long-range interaction of whole image and integrate spatial and spectral image information.
2) We design a spatial and spectral Mamba block to extract the spatial and spectral information, respectively. Benefiting from the strong long-range modeling capability of mamba, the proposed spatial and spectral Mamba block can model long-range interaction of whole image.
3) We propose a SSFM, which can adaptively estimate the importance of spatial and spectral information to guide their fusion. Besides, the residual learning idea is introduced to help with the module training.

We conduct extensive experiments on four diverse real-world HSI datasets. The results show that our method achieves superior performance, surpassing the state-of-the-art (SOTA) CNN-based and Transformer-based HSI classification models.

II. Related Work

A. HSI Classification

HSI classification methods typically fall into ML-based [10], [30] and DL-based [13], [15]. DL-based be roughly divided into GCN-based [11], [13], CNN-based [15], [31], and Transformer-based [32], [33] methods.

1) ML-Based: The early approaches adopt traditional machine learning, such as random forest (RF) [10], and support vector machine (SVM) [9], to classify the spectral features. However, the performance of these methods is limited because of lacking surrounding information. That is, the above methods utilize only the spectral information of a single pixel, neglecting the spatial information from surrounding neighboring pixels, which leads to suboptimal performance [3]. To remedy this problem, spatial information is incorporated into consideration. For example, the superpixel segmentation [34], the extended morphological profiles [35], and multiple kernel learning [36] are adopted to generate discriminative spectral–spatial features for classification. However, these methods mainly classify image using shallow features, so that the model cannot characterize the essential attribute of the objects [37]. This hinders the performance of classification.

2) GCN-Based: GCNs can be used to model long-range spatial relations in the HS image [11]. Qin et al. [38] simultaneously consider spatial and spectral neighborhoods by extending original GCN to second-order GCN. Wan et al. [39] utilized superpixel segmentation on HS image and fed it into GCN. The GiGCN model [40] from a superpixel viewpoint can fully exploit information inside and outside superpixels. Dynamic multiscale graph convolutional network classifier (DMSGer) simultaneously captures pixel-level and region-level features to boost classification performance [13]. Due to the GCNs' limitations in the graph construction, particularly for large graphs (need expensive computational cost), GCNs fail to classify or identify materials in large-scale hyperspectral scenes [41].

3) CNN-Based: CNN can automatically capture features from shallow to deep level by stacking layer by layer, thus attracting much attention in the field of HSI classification [42]. Hu et al. [43] extracts spectral features via a five-layer 1-D CNN. Considering the differences in scale of ground objects, [44] generates multiscale image patches for each pixel, and then extracts the multiscale features of the generated multiscale patches for classification. Furthermore, [45] jointly captures the spatial and spectral features via 3D-CNN. Different from the above patch-level methods, many image-level methods have been proposed to reduce running

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

time. SSFCN [3] is the first image-level HSI classification method, which is a double-branch FCN network to separately extract spectral and spatial features. Next, multiscale strategy and attention mechanism are introduced into FCN, which improve the multiscale feature extraction and fusion capabilities [46]. However, these CNN-based methods cannot model long-range dependencies due to the inherent locality of CNN, which hinders the feature extraction ability of models [19].

4) Transformer-Based: Recently, the Transformer have been widely used in HSI classification because of their abilities to model long-range dependencies and extract global spatial features. SpectralFormer [16] learns sequence information from local signatures with a transformer. Nevertheless, SpectralFormer ignores the spatial location information. To this end, some spatial–spectral transformers [32], [47] have been proposed to simultaneously capture spectral and spatial information. Unfortunately, these methods do not consider the differences in scales of different ground objects, limiting the ability to identify multiscale objects. In [42], a center-to-surrounding interactive learning (CSIL) framework based on the Transformer has been proposed to extract multiscale features for classification. Zhao et al. [12] pointed out that the transformer ignores the effective representation of local features due to the inherent modeling global correlation, and then the GSC-ViT proposed in [12] can capture local spectral–spatial information in HSI. Xue et al. [48] proposed a local transformer with spatial partition restore network (SPRLT-Net) to model locally detailed spatial discrepancies. Li et al. [49] proposed a lightweight Transformer architecture model, SPFormer, for few-shot HSI classification, which effectively reduces the parameters of transformer. However, the complexity of the self-attention mechanism is quadratic [18]. This poses challenges in terms of speed and memory usage, and hinders the model's ability to model long-range dependencies [19].

In general, existing HSI classification methods have limited ability to model the long-range dependencies because of its inherent properties, such as the locality of CNN and the quadratic complexity. Different from existing HSI classification methods, MambaHSI is the first Mamba-based HSI classification method that takes whole HSI image as the model's input, which can model long-range dependencies while maintaining linear computational complexity.

B. State-Space Models

Recent research advancements have sparked a surge of interest in the SSM. SSM originates from the classic Kalman Filter model, and modern SSMs are good at capturing long-range dependencies and can efficiently parallelize calculations [19]. SSMs are a novel alternative to CNNs or Transformers.

1) SSMs for Long Sequence: The Structured State-Space Sequence (S4) [22] model can model long-range dependencies. The promising property of linear complexity for sequence length attracts further exploration. Smith et al. [50] designed a S5 layer by introducing MIMO SSM and efficient parallel scan into S4 layer. Fu et al. [51] filled the performance gap between SSMs and Transformers in language modeling. Mehta et al. [52] introduced more gating units to improve the expressivity of S4. Recently, Mamba outperforms Transformers at various large-scale real data and maintains linear scaling in sequence length [24].

2) SSMs for Visual Applications: Nguyen et al. [53] extended 1-D S4 to handle 2-D images and 3-D videos. TranS4mer [54] combined the strengths of S4 and self-attention, and achieves SOTA performance for movie scene detection. Wang et al. [55] introduced a selectivity mechanism to S4, significantly improving the performance of S4 on long-form video understanding with lower memory footprint. U-Mamba [25] combined Mamba with U-shaped architecture for biomedical image segmentation.

3) SSMs for HSI Classification: Recently, [56], [57], [58], [59] adopt Mamba for hyperspectral image classification. SpectralMamba [56] introduced PSS and GSSM module to ease the sequentially learning in the state domain and rectify the spectrum, respectively. Huang et al. [58] proposed a spectral–spatial Mamba (3DSS-Mamba) for HSI classification. He et al. [57] explored the application of Mamba to hyperspectral blocks in 3-D from a 3-D perspective. However, the aforementioned methods utilize hyperspectral patches as input. This results in a significant amount of redundant computation when inferring the entire image, impeding practical deployment and application. Additionally, the fixed size of the patch inputs restricts the model's ability to fully exploit the image information, thereby limiting its feature extraction capacity.

To this end, we first design a spatial and spectral Mamba block to capture spatial and spectral information, respectively. Besides, we propose a SSFM to adaptively fuse the spatial and spectral information. This enables the proposed MambaHSI to fully integrate both spatial and spectral information while modeling long-range dependencies in a linear complexity manner.

III. Preliminaries

The modern SSM-based models, i.e., structure state-space sequence models (S4) and Mamba, both are inspired by a classical continuous system that maps a 1-D function or sequence, denoted as $x(t) \in \mathcal{R}$, to an output $y(t) \in \mathcal{R}$ through a hidden state $h(t) \in \mathcal{R}^N$. The process can be represented as a linear ordinary differential equation (ODE)

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t), \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathcal{R}^{N\times N}$ denotes the state matrix, and $\mathbf{B} \in \mathcal{R}^{N\times 1}$ and $\mathbf{C} \in \mathcal{R}^{N\times 1}$ signify the projection parameters.

The S4 and Mamba discretize the continuous system to make it more suitable for deep learning scenarios. Specifically, they introduce a timescale parameter $\mathbf{\Delta}$ and transform the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into discrete parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ using a fixed discretization rule. The commonly used discretization rule is zero-order hold (ZOH), which can be defined as follows:

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\mathbf{\Delta A}), \\ \overline{\mathbf{B}} &= (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A}) - \mathbf{I}) \cdot \mathbf{\Delta B}. \end{aligned} \tag{2}$$

After the discretization of $\overline{\mathbf{A}}, \overline{\mathbf{B}}$, the SSM-based models can be computed in two ways: linear recurrence or global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

convolution, defined as the following equations:

$$h_t = \overline{\mathbf{A}} h_{t-1} + \overline{\mathbf{B}} x_t, \quad y_t = \mathbf{C} h_t \tag{3}$$

$$\overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right), \quad y = x * \overline{\mathbf{K}} \tag{4}$$

where L and $\overline{\mathbf{K}} \in \mathcal{R}^L$ denote the length of the input sequence x and a structured convolutional kernel, respectively.

IV. Methodology

A. Motivation and Overall Idea

HSI classification is a pixel-level classification task [7]. This means that HSI classification models need to learn features that can finely distinguish subtle differences between pixels. Inspired by [60], the long-range modeling capability is vital for capturing discriminative features. Similar observations have been made in the HSI classification field [61], [62], [63]. Unfortunately, existing HSI classification methods have limited ability to model long-range dependencies due to inherent limitations, such as the locality of CNNs and the quadratic computational complexity of Transformers. Because of the quadratic computational complexity of Transformers, models based on Transformers often use patches as the basic token units when processing images, which hinders the model's ability to extract finer pixel-level representations for classification. This motivates us to develop a new framework that can model long-range dependencies with linear complexity, enabling finer feature extraction at the pixel level for classification. Therefore, we introduce Mamba as the basic unit to model long-range dependencies, which is vital for obtaining discriminative pixel-wise spatial features. However, Mamba ignores the fact that hyperspectral (HS) images are characterized by approximately contiguous spectral information. Therefore, we rethink HSI classification from a sequential perspective with Mamba and propose a spectral Mamba block to utilize the sequential nature of the spectrum. To comprehensively capture the spatial–spectral information, we propose a SSFM to adaptively fuse the spatial and spectral information, improving the recognizability of the extracted features.

B. Overview

An overview of the proposed MambaHSI is shown in Fig. 2. The framework contains three main components: embedding layer, encoder backbone, and segmentation head.

Embedding layer projects the spectral vector into an embedding space. Notably, different from the existing embedding methods based on patches, the embedding layer extracts embeddings of each pixel. This enables our model to obtain more fine-grained pixel embeddings, which is more suitable for dense prediction tasks such as hyperspectral classification. Specifically, the fine-grained pixel embedding $\mathbf{E} \in R^{H\times W\times D}$ can be obtained from HSI $\mathbf{I} \in R^{H\times W\times C}$ as follows:

$$\begin{aligned}\mathbf{E} &= \text{Embedding}(\mathbf{I}) \\ &= \text{SiLU}(\text{GN}(\text{Conv}(\mathbf{I})))\end{aligned} \tag{5}$$

where Conv, GN, and SiLU denote the convolutional layer with 1×1 kernel size, the group norm (GN) layer, and SiLU activation function, respectively. H, W, and C signify the height, width, and spectral channel number of the input HSI, respectively. D is the embedding dimension. $\mathbf{I}$ and $\mathbf{E}$ denote the input HSI and the extracted embedding.

Encoder backbone is used to extract the discriminative spatial–spectral features for classification. Specifically, the encoder backbone mainly contains three components: spatial Mamba block (SpaMB) for extracting spatial features, spectral Mamba block for capturing spectral features, and SSFM for integrating spatial and spectral features. The process can be defined as follows:

$$\mathbf{H} = \text{Encoder}(\mathbf{E}) \tag{6}$$

where Encoder and $\mathbf{H}$ denote the encoder backbone and the extracted hidden features.

Segmentation head adopts convolutional layer with 1×1 kernel size to obtain the final logits l, i.e., $\mathbf{l} = \text{SegHead}(\mathbf{H})$.

C. Spatial Mamba Block

The hyperspectral classification is a pixel-level classification task. This means that the representation used for classification needs to satisfy two conditions. First, the representation should be refined and should reflect the differences between pixels. Therefore, different from the existing patch methods, we extract embedding in a pixel-level manner. Second, the representations should be discriminative for classification. Hence, we require that the proposed modules have strong long-distance modeling capabilities. Notably, the transformer has quadratic computational complexity and cannot be used to establish long-distance dependencies at the pixel level. This compels us to design a novel spatial feature extractor to build long-range dependencies in linear computational complexity. In this article, we adopt the Mamba layer as the basic unit to build the spatial feature extractor, which can model long-range dependencies in linear computational complexity.

The detailed structure of the SpaMB is shown in Fig. 2. The forward process can be formulated as follows:

$$\begin{aligned}\mathbf{HF_{spa}} &= \text{Flatten}\left(\mathbf{H^i}\right), \\ \mathbf{HR_{spa}} &= \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\mathbf{HF_{spa}}\right)\right)\right), \\ \mathbf{H^o_{spa}} &= \text{Reshape}\left(\mathbf{HR_{spa}}\right) + \mathbf{H^i}\end{aligned} \tag{7}$$

where $\mathbf{H^i} \in \mathcal{R}^{B\times H\times W\times D}$ and $\mathbf{H^o_{spa}} \in \mathcal{R}^{B\times H\times W\times D}$ denote the input embeddings of pixel-level and the output features of SpaMB. B, H, W, and D signify the batch size, image height, image width, and embedding dimension, respectively. The embedding dimension D is set to 128 in experiments. $\mathbf{HF_{spa}} \in \mathcal{R}^{B\times L1\times D}$ and $\mathbf{HR_{spa}} \in \mathcal{R}^{B\times L1\times D}$ denote the flatten input and the learned residual spatial feature. $L1$ is equal to $H \times W$. The Mamba denotes the standard Mamba block proposed in [24]. The design of GN and residual connection aid the SpaMB learning.

D. Spectral Mamba Block

Unlike traditional visual systems that capture image through RGB channels, HSIs can cover a much larger spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

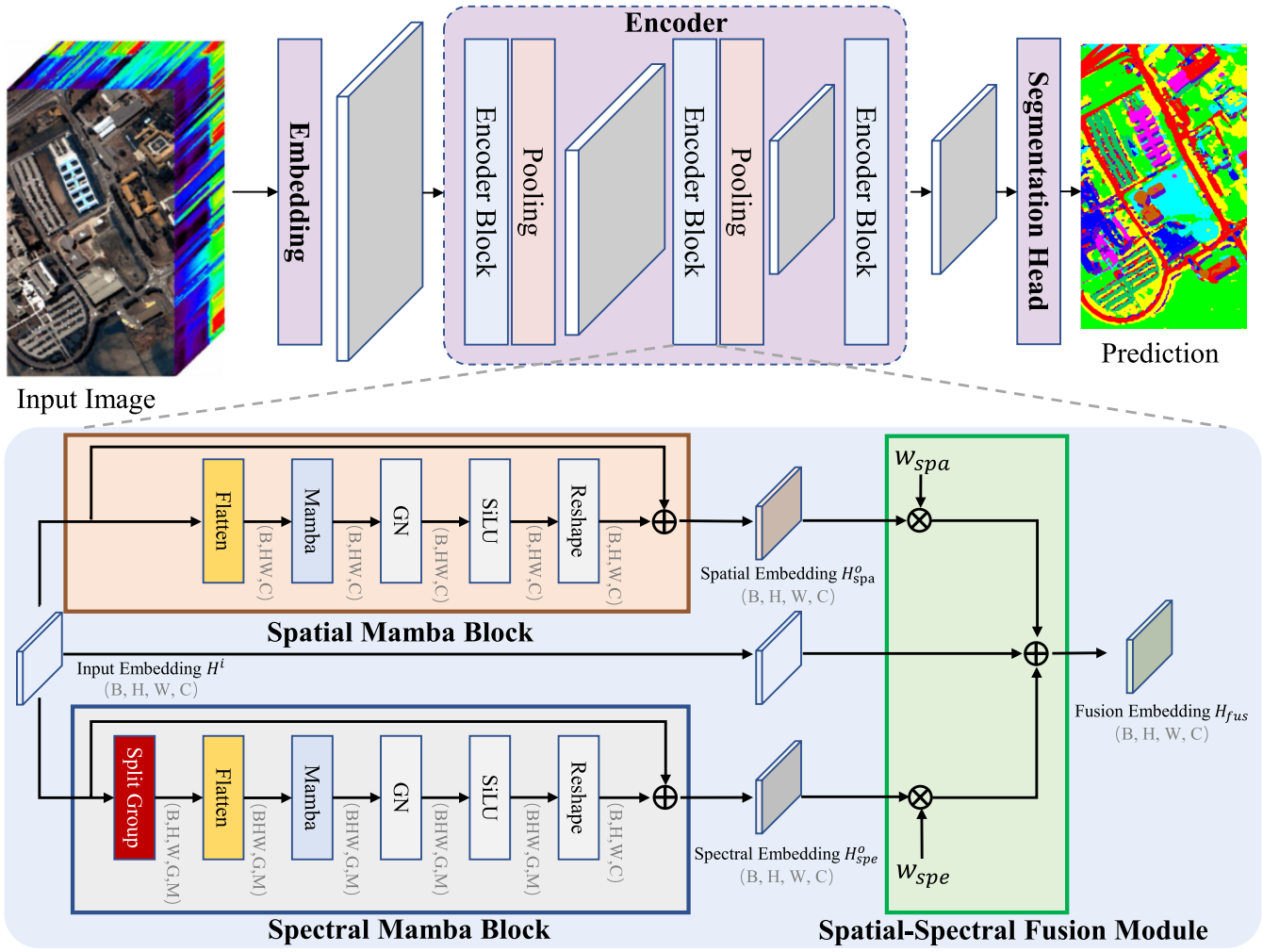


Fig. 2. Overview of the proposed MambaHSI framework. The whole HSI is fed into the embedding layer to obtain pixel-level embeddings. Then, these embeddings are taken as the inputs of the encoder to model the long-range dependencies and capture the discriminative features. Finally, the segmentation head classifies the features extracted by the encoder to obtain the final prediction. The encoder block contains three components: SpaMB for capturing spatial features, spectral Mamba block for extracting spectral features, and SSFM to fuse the spatial and spectral features.

range and higher spectral resolution. How to model the relationship between spectra and extract discriminative features is still an open research problem. In this article, we design a SpeMB to achieve the above goal. The overall structure is shown in Fig. 2. Specifically, we divide a spectral features into G group. Then, we model the relations between different spectral groups, and then update the spectral features guided by the mined relationships between spectral groups. The extracted spectral features H^o can be obtained as follows:

$$\begin{aligned}\mathbf{HG_{spe}} &= \text{SplitSpectralGroup}\left(\mathbf{H^i}\right),\\ \mathbf{HF_{spe}} &= \text{Flatten}\left(\mathbf{HG_{spe}}\right),\\ \mathbf{HR_{spe}} &= \text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\mathbf{HF_{spe}}\right)\right)\right),\\ \mathbf{H^o_{spe}} &= \text{Reshape}\left(\mathbf{HR_{spe}}\right) + \mathbf{H^i} \end{aligned} \tag{8}$$

where $\mathbf{HG_{spe}} \in \mathcal{R}^{B\times H\times W\times G\times M}$, $\mathbf{HF_{spe}} \in \mathcal{R}^{N\times G\times M}$, $\mathbf{HR_{spe}} \in \mathcal{R}^{N\times G\times M}$, and $\mathbf{H^o_{spe}} \in \mathcal{R}^{B\times H\times W\times D}$ denote the divided spectral group features, the flatten ones, the residual ones, and the output spectral features, respectively. G represents the number of groups into which the semantic vector is split. M equals the pixel embedding dimension D divided by the group number G. N is equal to $B\times H\times W$. The Mamba signifies the standard mamba block proposed in [24].

E. Spatial–Spectral Fusion Module

Both spatial and spectral features are vital for HSI classification, and integrating spatial and spectral information is beneficial for classification [3]. This motivates us to design a SSFM, and the architecture of SSFM is shown in Fig. 2. Considering that hyperspectral classification usually has fewer labeled samples, we introduced the idea of residual learning to alleviate overfitting that may occur in the training process. Besides, the SSFM adaptively estimates the importance of spatial and spectral to guide the fusion. The fusion process can be formulated as follows:

$$\mathbf{H_{fus}} = \mathbf{H_i} + w_{\text{spa}} \times \mathbf{H^o_{spa}} + w_{\text{spe}} \times \mathbf{H^o_{spe}} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Training and Test Procedures of MambaHSI

```
// Training
Input: Whole image I and training label Y_tr
Initialize: Initialize network parameters Θ with random values
for e = 1, 2, ..., E do
    // Whole Image Forward Process
    E = Embedding(I); //Extract Image Embedding
    H = Encoder(E); //Encoder incudes SpaMB, SpeMB, and
    SSFM modules.
    l = SegHead(H); //Obtain Image Logits
    // Compute Loss
    L ← {l, Y_tr^s} in Eq.(10).
    // Update Parameters
    Update the parameters of MambaHSI using Adam.
end
Output: Network parameters Θ
// Testing
Input: Whole test image I_te
1) Load trained network parameter Θ
2) predict the testing image with the trained model
Output: The class prediction ŷ
```

where w_{spa} and w_{spe} denote the fusion weight of spatial and spectral, respectively. w_{spa} and w_{spe} are randomly initialized, and these weights are updated through backpropagation to determine the final fusion weight.

F. Computational Complexity

Considering that the number of pixels L is much larger than the number of images B, the number of feature channels D and other parameters, we analyzed the relationship between the pixel number L and computational complexity. The computational complexity of a Transformer is quadratic with respect to the length of input sequence [18], i.e., $O(L^2)$. In addition, Mamba has a linear complexity with respect to the sequence length [24], i.e., $O(L)$. The proposed MambaHSI consists of embedding layer, SpaMB, spectral mamba block, SSFM, and segmentation head. Their complexity all are $O(L)$. Therefore, the complexity of the proposed MambaHSI is $O(L)$

G. Training and Inference

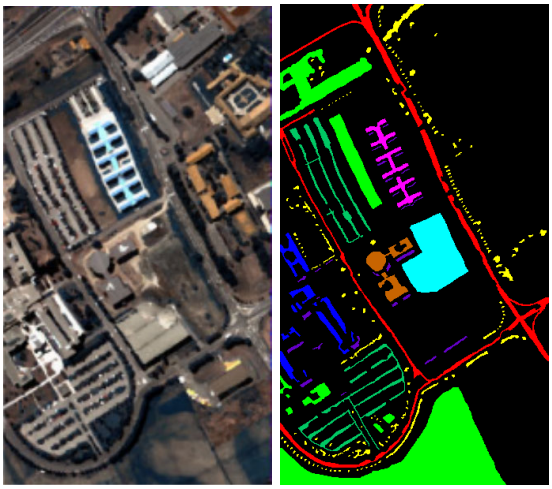
The training and test procedure is presented in Algorithm 1. Unlike patch-level HSI classification methods [12], [15], MambaHSI is an image-level HSI method, which adopts the whole image as input and can obtain all pixel's prediction of whole image by one time forward. Additionally, the encoder, including the SpaMB, SpeMB, and SSFM, extracts the spatial–spectral features. MambaHSI can be trained in an end-to-end manner. In training phase, the training loss function $\mathcal{L}$ is given as

$$\mathcal{L} = \text{CrossEntropy}(\mathbf{l}, \mathbf{Y_{tr}}) \tag{10}$$

where $\mathbf{Y_{tr}}$ is the training label and $\mathbf{l} = \text{MambaHSI}(\mathbf{I})$ is the logits output by MambaHSI. $\mathbf{I}$ is the whole image.

In inference phase, given a test image ($\mathbf{I_{te}}$), the final prediction $\hat{y}$ can be obtained as follows:

$$\hat{y} = \arg\max(\text{softmax}(\text{MambaHSI}(\mathbf{I_{te}}))). \tag{11}$$



(a) (b)

ID	Color	Category	Train	validation	Test	Total
1		Asphalt	30	10	6591	6631
2		Meadows	30	10	18609	18649
3		Gravel	30	10	2059	2099
4		Trees	30	10	3024	3064
5		Metal sheets	30	10	1305	1345
6		Bare soil	30	10	4989	5029
7		Bitumen	30	10	1290	1330
8		Bricks	30	10	3642	3682
9		Shadows	30	10	907	947
Total			270	90	42416	42776

(c)

Fig. 3. Pavia University dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

V. Experiments

In this section, we first describe the experimental setup, including datasets, evaluation metrics, comparison methods, and implementation details. Then, we comprehensively compare with SOTAs methods both qualitatively and quantitatively. Finally, we conduct a detailed ablation study to analyze the effect of the proposed modules.

A. Experimental Setup

Datasets: To more comprehensively evaluate the effectiveness of the proposed model, we choose four widely used and diverse hyperspectral datasets, i.e., Pavia University (PaviaU), Houston, WHU-Hi-HanChuan (HanChuan) [64], and WHU-Hi-HongHu (HongHu) [64].

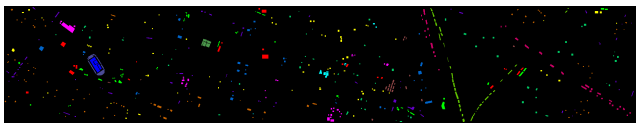
1) Pavia University: The image was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia University, consisting 103 bands with a spatial size of 610 × 340. The dataset contains 42776 labeled pixels of nine classes. The related information, such as original image visualization, ground truth map, and the dataset split configuration, is shown in Fig. 3.

2) Houston: The image was acquired over the University of Houston campus and its neighbor regions by ITRES CASI 1500 HS imager with 144 spectral bands, and has an image size of 349 × 1905. It was provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest. The dataset contains 16 classes, including residual, commercial, and so on. More details are shown in Fig. 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

ID	Color	Category	Train	validation	Test	Total
1		Healthy Grass	30	10	1211	1251
2		Stressed Grass	30	10	1214	1254
3		Synthetic Grass	30	10	657	697
4		Tree	30	10	1204	1244
5		Soil	30	10	1202	1242
6		Water	30	10	285	325
7		Residential	30	10	1228	1268
8		Commercial	30	10	1204	1244
9		Road	30	10	1212	1252
10		Highway	30	10	1187	1227
11		Railway	30	10	1195	1235
12		Parking Lot 1	30	10	1193	1233
13		Parking Lot 2	30	10	429	469
14		Tennis Court	30	10	388	428
15		Running Track	30	10	620	660
Total			450	150	14429	15029

(c)

Fig. 4. Houston dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.



(a)



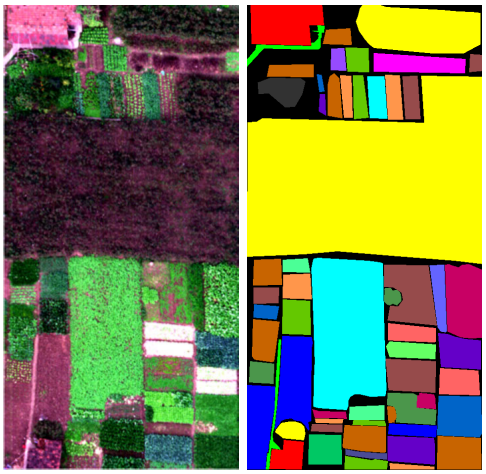
(b)

ID	Color	Category	Train	validation	Test	Total
1		Strawberry	30	10	44695	44735
2		Cowpea	30	10	22713	22753
3		Soybean	30	10	10247	10287
4		Sorghum	30	10	5313	5353
5		Water spinach	30	10	1160	1200
6		Watermelon	30	10	4493	4533
7		Greens	30	10	5863	5903
8		Trees	30	10	17938	17978
9		Grass	30	10	9429	9469
10		Red roof	30	10	10476	10516
11		Gray roof	30	10	16871	16911
12		Plastic	30	10	3639	3679
13		Bare soil	30	10	9076	9116
14		Road	30	10	18520	18560
15		Bright object	30	10	1096	1136
16		Water	30	10	75361	75401
Total			480	160	256890	257530

(c)

Fig. 5. HanChuan dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

3) HanChuan: The WHU-Hi-HanChuan dataset was acquired from 17:57 to 18:46 on June 17, 2016, in Hanchuan, Hubei province, China, with an 17 mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform. The study area is a rural–urban fringe zone with buildings, water, and cultivated land, which contains seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The size of the imagery is 1217 × 303 pixels, and there are 274 bands from 400 to 1000 nm. An overview of this dataset is given in Fig. 5.



(a) (b)

ID	Color	Category	Train	validation	Test	Total
1		Red roof	30	10	14001	14041
2		Road	30	10	3472	3512
3		Bare soil	30	10	21781	21821
4		Cotton	30	10	163245	163285
5		Cotton firewood	30	10	6178	6218
6		Rape	30	10	44517	44557
7		Chinese cabbage	30	10	24063	24103
8		Pakchoi	30	10	4014	4054
9		Cabbage	30	10	10779	10819
10		Tuber mustard	30	10	12354	12394
11		Brassica parachinensis	30	10	10975	11015
12		Brassica chinensis	30	10	8914	8954
13		Small Brassica chinensis	30	10	22467	22507
14		Lactuca sativa	30	10	7316	7356
15		Celtuce	30	10	962	1002
16		Film covered lettuce	30	10	7222	7262
17		Romaine lettuce	30	10	2970	3010
18		Carrot	30	10	3177	3217
19		White radish	30	10	8672	8712
20		Garlic sprout	30	10	3446	3486
21		Broad bean	30	10	1288	1328
22		Tree	30	10	4000	4040
Total			660	220	385813	386693

(c)

Fig. 6. HongHu dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

4) HongHu: The image was acquired from 16:23 to 17:37 on November 20, 2017, in Honghu City, Hubei province, China, with a 17 mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform. The experimental area is a complex agricultural scene with many classes of crops, and different cultivars of the same crop are also planted in the region. The size of the imagery is 940 × 475 pixels, and there are 270 bands from 400 to 1000 nm. More information is given in Fig. 6.

Evaluation Metrics: Three evaluation metrics widely used in the hyperspectral classification were adopted in the experiments, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). To eliminate the deviation caused by randomly selecting training samples, the main comparative experiments and ablation studies were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULT (ACC% ± STD%) OF PAVIA UNIVERSITY. BEST IN **BOLD** AND SECOND WITH UNDERLINE. THESE NOTES ARE THE SAME TO OTHERS

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI
	TGRS2004	TGRS2005	TNNLS2022	TNNLS	TGRS2022	TGRS2024	TGRS2021	TGRS2024	Ours
Asphalt	93.69±2.34	93.74±1.91	81.49±4.78	90.36±1.73	90.98±7.20	**98.27±1.00**	92.60±2.29	97.10±1.64	94.35±1.73
Meadows	88.56±2.41	87.29±1.93	87.60±3.00	98.86±0.78	89.87±13.81	**98.98±0.76**	93.58±2.14	97.68±1.32	96.43±1.92
Gravel	50.42±5.13	48.47±4.74	**97.15±1.28**	91.66±6.33	91.75±5.61	87.83±7.54	60.90±8.15	86.46±7.42	93.39±5.35
Trees	55.88±5.67	55.21±4.85	87.68±2.85	81.34±10.11	93.50±6.74	**97.80±1.46**	92.28±6.23	91.83±9.94	89.75±3.60
Metal sheets	95.67±1.95	92.06±1.78	99.88±1.68	91.15±1.31	99.54±0.60	99.68±0.59	98.00±1.72	99.45±0.53	**99.99±0.02**
Bare soil	49.11±8.63	44.29±5.43	**99.66±6.60**	98.87±1.90	92.36±9.33	85.27±7.50	63.92±12.41	79.65±9.24	98.60±0.93
Bitumen	41.46±3.09	48.42±4.67	**99.36±0.88**	99.28±0.61	96.34±6.58	85.92±7.00	48.11±8.47	86.63±10.10	96.50±3.13
Bricks	73.17±4.61	70.94±6.73	88.65±2.94	87.68±7.46	89.67±4.82	90.93±4.12	81.22±4.66	89.50±4.70	**94.40±2.93**
Shadows	**99.91±0.08**	99.81±0.11	96.89±0.82	97.21±3.36	96.73±9.02	97.12±2.03	88.78±5.92	98.71±1.10	99.36±0.62
OA(%)	71.31±3.15	70.68±2.61	89.55±1.22	94.47±0.97	91.31±9.36	94.97±1.43	91.10±2.60	92.65±2.30	**95.74±0.90**
AA(%)	71.98±1.39	71.14±1.61	93.15±0.70	92.93±1.07	93.41±6.57	93.53±1.59	79.93±2.67	91.89±1.73	**95.86±1.11**
Kappa(%)	63.96±3.19	63.13±2.81	86.49±1.54	92.67±1.26	93.12±5.19	93.38±1.84	77.40±4.41	90.38±2.95	**95.00±2.24**

TABLE II

QUANTITATIVE RESULT (ACC% ± STD%) OF HOUSTON DATASET

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI
	TGRS2004	TGRS2005	TNNLS2022	TNNLS2022	TGRS2022	TGRS2024	TGRS2021	TGRS2024	Ours
Healthy Grass	93.68±5.88	93.00±0.45	91.27±4.59	83.21±5.03	92.69±7.57	88.24±9.00	91.29±5.38	90.16±6.48	**95.20±4.66**
Stressed Grass	85.92±6.74	91.05±2.57	91.29±4.73	82.83±7.41	85.06±16.86	93.41±5.99	92.12±6.99	95.77±2.95	**98.29±1.26**
Synthetic Grass	98.83±1.38	97.82±3.65	99.62±0.39	98.83±1.66	99.21±0.74	95.21±9.84	94.68±3.25	**99.83±0.26**	99.74±0.46
Tree	**98.89±0.50**	96.88±3.62	83.67±4.98	72.42±10.94	93.11±3.19	97.44±3.80	93.32±9.22	94.78±7.69	97.26±2.47
Soil	90.87±2.41	87.75±4.04	98.76±1.79	93.14±4.62	97.07±6.99	97.07±2.86	98.18±1.72	97.63±2.69	**99.54±0.76**
Water	87.86±14.30	87.30±0.68	**99.79±0.63**	88.99±6.78	97.58±1.78	94.85±4.72	87.21±8.11	96.03±2.59	97.47±2.33
Residential	73.99±4.46	74.97±4.97	85.21±3.03	82.34±5.24	88.62±2.41	**94.02±2.94**	91.47±3.54	93.11±2.39	93.03±2.25
Commercial	79.40±11.34	83.26±3.92	67.40±5.02	91.02±6.25	76.74±9.92	**93.80±3.94**	84.41±7.91	90.53±8.57	81.35±3.98
Road	65.11±1.49	68.92±3.43	84.79±3.03	85.51±6.43	83.11±5.86	88.32±2.83	87.75±3.83	85.20±4.96	**90.03±3.40**
Highway	70.50±4.76	73.48±6.01	**98.16±1.89**	94.92±2.69	91.88±9.34	76.55±6.96	82.09±5.12	85.28±4.22	96.55±1.49
Railway	70.15±2.45	66.13±3.26	94.22±4.37	**96.52±3.78**	90.33±5.02	91.57±4.40	85.90±4.63	93.08±3.33	92.74±2.18
Parking Lot 1	64.23±4.99	58.37±5.03	91.48±4.79	**92.45±3.63**	83.43±9.23	83.37±6.17	85.33±5.87	88.46±3.14	91.19±3.68
Parking Lot 2	43.46±10.62	31.71±4.80	96.95±2.60	88.10±5.13	92.98±4.32	89.11±4.64	86.81±6.28	90.01±4.69	**97.67±2.04**
Tennis Court	88.44±5.61	80.89±5.68	**100.00±0.00**	98.30±2.67	**100.00±0.00**	97.98±2.63	91.65±6.36	97.01±5.11	**100.00±0.00**
Running Track	99.16±0.46	97.08±1.84	99.97±0.10	99.14±1.89	98.47±3.70	96.08±3.96	96.18±2.66	97.73±1.23	**100.00±0.00**
OA(%)	79.63±1.04	78.95±0.90	90.36±1.29	88.13±2.26	89.79±4.20	90.31±1.73	89.13±2.12	91.85±1.05	**94.46±0.83**
AA(%)	80.70±1.03	79.24±0.83	92.17±1.07	89.85±1.91	91.35±3.55	91.80±1.46	89.89±1.74	92.97±0.92	**95.34±0.78**
Kappa(%)	77.97±1.12	77.25±0.98	89.58±1.39	87.16±2.44	90.64±4.68	89.52±1.87	88.24±2.29	91.18±1.14	**94.21±1.93**

TABLE III

QUANTITATIVE RESULT (ACC% ± STD%) OF HANCHUAN DATASET

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI
	TGRS2004	TGRS2005	TNNLS2022	TNNLS2022	TGRS2022	TGRS2024	TGRS2021	TGRS2024	Ours
Strawberry	92.03±1.24	87.87±1.73	74.51±5.53	89.34±2.23	78.59±12.03	**95.97±1.87**	95.13±1.25	95.26±1.50	89.72±5.75
Cowpea	69.50±7.79	57.02±4.42	62.51±8.18	81.64±3.38	84.21±4.34	**92.48±2.56**	87.27±3.37	91.22±3.05	81.37±5.69
Soybean	47.41±4.94	45.31±2.24	94.63±2.80	81.06±6.56	79.85±9.55	80.03±9.18	70.36±6.15	78.37±7.47	**94.88±3.73**
Sorghum	74.49±9.57	51.50±2.59	**99.51±0.50**	67.09±4.82	84.68±13.06	91.12±7.72	88.62±6.67	85.17±12.38	98.46±1.35
Water spinach	9.72±0.77	9.25±0.78	99.18±0.92	34.98±2.76	68.38±27.44	56.82±12.89	42.01±11.48	46.57±13.03	**99.54±0.85**
Watermelon	15.28±2.00	11.66±1.03	**94.18±2.43**	53.81±9.81	9.39±15.90	38.07±8.70	34.68±6.40	35.97±7.01	76.99±4.95
Greens	36.69±2.30	45.58±3.46	**98.72±1.26**	80.84±8.87	92.94±4.47	68.87±17.46	67.69±10.29	66.90±15.67	92.61±4.63
Trees	65.71±5.51	66.22±4.84	84.57±4.09	89.27±4.80	**91.24±1.96**	81.52±7.87	83.17±4.16	82.39±5.23	70.73±7.10
Grass	47.93±7.71	40.87±7.40	72.25±5.64	65.14±6.79	70.48±8.43	68.59±6.61	63.00±4.53	60.64±17.62	**89.01±3.62**
Red roof	74.50±11.85	59.59±6.64	87.81±3.96	93.03±4.21	85.23±6.83	90.79±14.71	88.76±12.99	93.22±4.31	**94.44±2.18**
Gray roof	35.90±1.90	54.20±9.34	**96.13±1.66**	93.86±5.06	92.58±4.29	87.86±4.63	77.41±13.19	82.13±7.96	89.56±4.91
Plastic	13.49±2.57	17.50±2.66	**99.09±1.04**	89.22±7.45	39.55±27.62	63.48±14.80	51.40±6.89	51.92±6.01	88.83±6.91
Bare soil	27.36±3.66	33.89±4.49	**82.67±3.97**	66.66±6.60	82.37±2.46	50.21±6.71	47.62±9.17	53.28±10.88	78.69±3.73
Road	80.55±8.56	76.20±6.53	75.85±3.16	**91.25±2.43**	88.31±4.97	90.78±2.74	88.92±3.08	90.64±3.07	89.68±3.03
Bright object	33.15±10.97	26.80±10.82	**98.36±2.21**	46.57±17.42	97.30±2.18	62.42±10.67	52.51±8.59	70.05±12.35	94.42±2.41
Water	**99.95±0.06**	98.82±1.33	92.23±5.31	99.75±0.14	93.64±4.75	98.97±1.02	99.47±0.33	99.77±0.23	98.28±1.62
OA(%)	60.98±1.35	64.09±1.78	84.39±1.83	86.78±1.43	84.88±3.02	85.14±2.93	82.99±2.31	83.14±3.96	**90.21±1.67**
AA(%)	51.48±1.46	48.89±1.15	88.26±0.89	76.47±1.62	77.42±3.61	76.13±3.26	71.13±2.56	73.97±2.06	**89.20±1.36**
Kappa(%)	56.00±1.49	59.05±1.86	82.01±2.02	84.64±1.62	75.80±4.70	82.75±3.31	80.29±2.61	80.57±4.33	**89.07±2.31**

conducted ten times, while all other analytical experiments were performed five times. The larger the mean value and the smaller the standard deviation of these metrics, the better the performance of the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF HONGHU DATASET

Class	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM	RF	DMSGer	GiGCN	FullyContNet	CLOLN	Spectralformer	GSC-ViT	MambaHSI
	TGRS2004	TGRS2005	TNNLS2022	TNNLS2022	TGRS2022	TGRS2024	TGRS2021	TGRS2024	Ours
Red roof	94.30±1.90	84.35±5.74	92.96±4.02	90.93±2.64	96.78±1.57	95.36±5.56	96.66±1.63	**98.16±1.08**	96.47±1.50
Road	53.77±4.54	52.01±5.17	93.34±4.05	51.88±7.77	**95.63±2.54**	78.60±6.19	62.84±4.24	73.78±9.09	95.43±2.33
Bare soil	90.70±3.65	90.92±2.70	92.15±3.46	78.80±2.87	92.23±1.51	**98.22±1.32**	95.60±2.81	97.21±1.81	94.66±1.05
Cotton	97.42±0.34	97.09±0.51	60.26±10.53	99.37±0.32	93.59±2.92	99.48±0.37	99.42±0.30	**99.57±0.14**	96.96±1.79
Cotton firewood	17.61±2.78	14.17±2.64	**99.30±0.79**	62.41±8.43	96.03±2.60	41.33±18.92	42.14±16.25	37.03±9.33	96.20±1.77
Rape	90.59±1.93	84.04±1.60	84.36±4.27	89.61±3.17	91.61±2.28	**96.09±1.67**	95.73±1.02	95.72±1.83	93.90±2.47
Chinese cabbage	78.60±3.21	72.03±4.93	69.44±3.44	82.91±3.07	82.52±3.43	**92.72±1.96**	90.05±2.26	90.38±3.66	86.60±3.30
Pakchoi	13.56±2.31	9.13±2.07	**98.36±2.87**	75.44±8.66	96.81±2.03	46.44±6.02	35.59±5.04	32.12±4.83	96.08±2.90
Cabbage	96.57±1.71	90.60±3.89	95.14±1.73	85.39±3.04	96.62±1.20	97.04±2.05	**97.89±1.07**	97.06±2.73	94.78±1.69
Tuber mustard	50.25±7.08	33.29±6.19	87.22±4.16	**92.42±2.80**	90.08±2.81	84.88±3.47	78.81±4.58	79.23±7.53	90.45±2.72
Brassica parachinensis	31.70±5.48	25.71±4.48	82.08±7.09	75.17±4.50	86.27±3.66	72.03±8.35	69.29±7.09	72.41±8.95	**91.96±2.87**
Brassica chinensis	42.58±4.37	40.77±3.36	**91.91±2.46**	82.52±5.36	88.62±4.09	66.63±8.82	58.68±5.96	65.61±8.66	87.40±5.22
Small Brassica chinensis	59.05±4.52	54.24±3.33	73.10±6.40	**89.28±2.70**	89.16±2.24	82.67±5.83	76.20±6.07	77.49±8.47	89.12±3.12
Lactuca sativa	67.70±6.77	57.66±7.93	**94.87±2.43**	68.03±5.44	93.88±1.93	87.59±7.03	83.89±6.15	86.26±11.93	94.01±1.65
Celtuce	8.86±4.25	3.78±1.16	**99.85±0.28**	21.65±2.65	98.42±1.16	75.88±12.93	61.24±13.37	63.62±20.25	98.84±1.17
Film covered lettuce	87.85±2.55	80.73±7.57	96.89±1.26	71.70±6.79	88.96±4.28	97.39±1.73	95.23±2.99	**97.42±1.66**	96.83±1.60
Romaine lettuce	55.60±4.87	54.14±6.89	**99.23±1.01**	54.94±4.04	95.10±4.52	80.09±9.70	68.68±8.79	84.92±6.09	98.60±1.49
Carrot	28.46±3.59	18.32±1.20	**98.44±2.49**	67.65±8.17	97.22±1.43	65.55±15.89	60.99±7.01	67.44±10.50	96.72±1.82
White radish	65.15±5.12	47.85±5.57	88.34±4.02	72.71±8.86	89.35±3.28	82.53±7.23	82.77±4.78	85.10±8.11	**91.30±3.07**
Garlic sprout	37.85±5.53	17.29±4.70	99.02±0.85	66.48±6.47	98.34±1.10	81.38±6.60	71.18±8.29	71.64±10.46	**99.18±0.42**
Broad bean	9.35±0.92	9.68±0.83	**99.96±0.12**	65.20±14.58	99.95±0.12	30.19±12.06	28.91±8.18	25.92±7.91	99.91±0.18
Tree	21.53±1.27	20.00±1.27	**99.88±0.24**	85.35±8.84	99.11±1.00	55.02±10.89	58.01±15.28	46.26±11.15	99.54±0.51
OA(%)	68.77±1.27	59.85±2.13	75.43±4.34	87.22±0.60	92.19±1.35	87.93±2.11	86.04±2.83	86.18±1.86	**94.58±1.01**
AA(%)	54.50±0.41	48.08±1.05	90.73±0.92	74.08±1.50	93.47±0.48	77.60±2.25	73.17±2.45	74.74±1.64	**94.77±0.61**
Kappa(%)	62.71±1.28	53.22±1.95	71.59±4.32	83.95±0.73	93.86±1.54	85.06±2.49	82.72±3.22	82.93±2.16	**94.00±1.88**

Comparison Methods: We compare relative HSI classification methods.

1) For ML Based:
   a) SVM[TGRS2004] [9]: The model adopts a SVM to address HSI classification.
   b) RF[TGRS2005] [10]: The model exploits RF to classify HSI data.
2) For GCN Based:
   a) DMSGer[TNNLS2022] [13]: The model utilizes a dynamic multiscale GCN classifier to capture pixel-level and region-level features.
   b) GiGCN[TNNLS2022] [40]: The model exploits information both inside and outside superpixels to boost classification performance.
3) For CNN Based:
   a) FullyContNet[TGRS22] [46]: The model combines a scale attention mechanism and a multiscale feature extractor, such as pyramid pooling, to obtain more effective contexts for classification. The input of this method is the original HSI.
   b) CLOLN[TGRS24] [15]: The model is a channel-layer-oriented lightweight network to alleviate the number of parameters and computational complexity associated with CNNs.
4) For Transformer Based:
   a) Spectralformer [TGRS21] [16]: The model extracts spectrally local sequence information from neighboring bands of HS images, yielding group-wise spectral embeddings.
   b) GSC-ViT[TGRS24] [12]: The model adopts a groupwise separable convolution ViT to capture local and global spectral–spatial information for HSI classification.

Implement Details: The proposed MambaHSI is implemented with pytorch. All experiments are randomly conducted ten times to eliminate the influence of random sample selection and parameter initialization. In each trial, the training and validation set are composed of 30 and 10 samples randomly selected from the images. The testing set is made up of the remaining samples. As the proposed MambaHSI takes the whole image as input and obtains the predictions at once, the batch size is set to 1. The Adam optimizer is used to train the model. The learning rate is set to 0.0003. The group number G and hidden dimension D are set to 4 and 128. The timescale parameter Δ, and projection parameters B and C in the mamba block follow the parameter settings of the Mamba [24]. All runs are conducted on the same hardware: NVIDIA GeForce RTX 3090 GPU, ×64 Intel Xeon Gold 6226R CPU, and 256 GB RAM.

B. Comparison With SOTAs

We comprehensively compare our method with several SOTA HSI classification methods from a quantitative and qualitative perspective. These comparison methods cover ML-based [9], [10], GCN-based [13], [40], CNN-based [15], [46], and Transformer-based [12], [16] methods. The quantitative and qualitative assessments are reported in Tables I–IV and Figs. 7–10, respectively. From the quantitative results, we find the similar observation with [42], i.e., the superiority of different backbones is not always prominent, especially in different datasets. Take the Houston dataset as an example, the best transformer-based method GSC-ViT outperforms the best CNN-based CLOLN by 1.54% in OA. In contrast, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

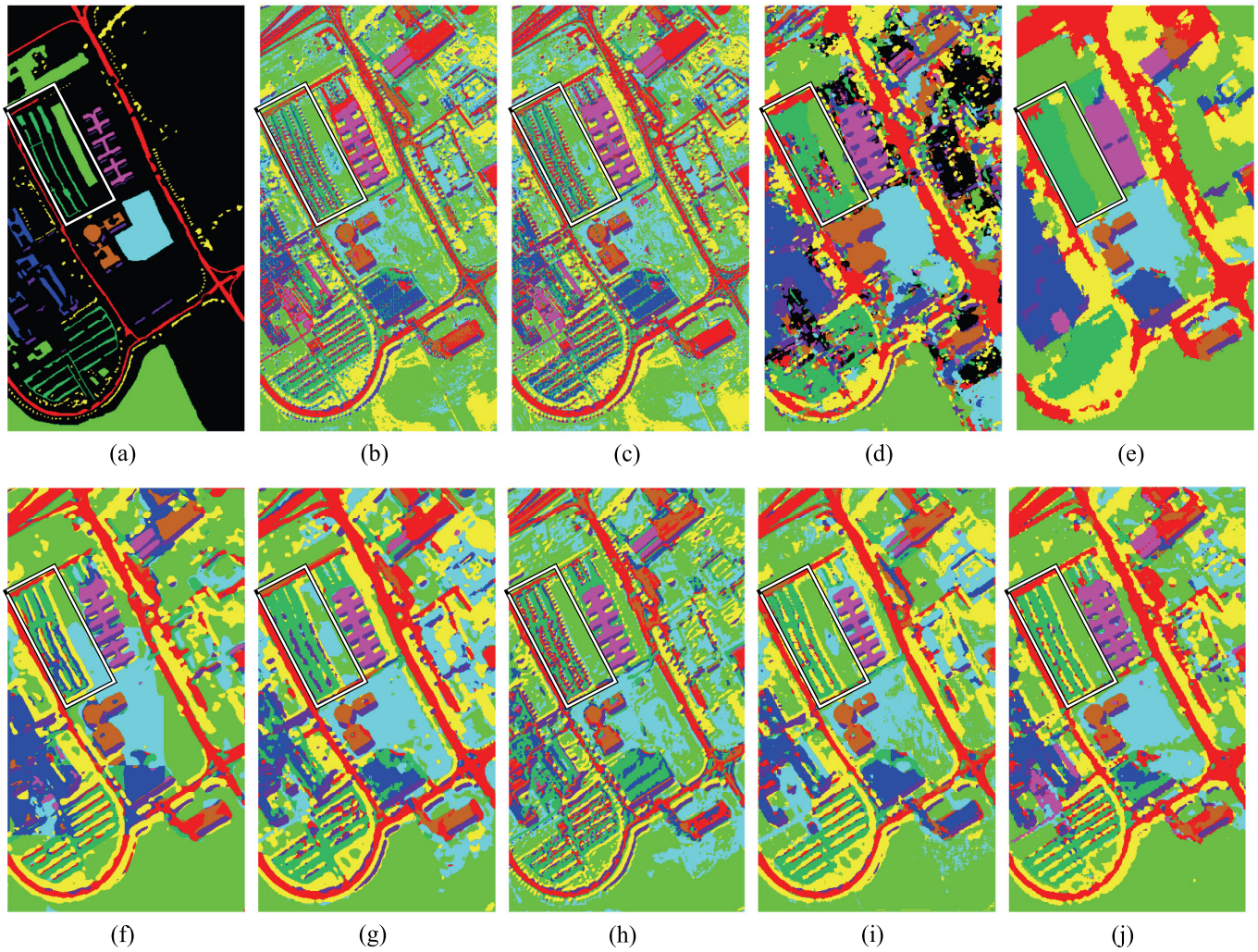


Fig. 7. Qualitative visualization of the classification map for Pavia University dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) DMSGer. (e) GiGCN. (f) FullyContNet. (g) CLOLN. (h) Spectralformer. (i) GSC-ViT. (j) Proposed MambaHSI. The meaning of colors refers to Fig. 3(c).

best CNN-based CLOLN gains about 2.32% improvement in OA over the best transformer-based GSC-ViT on Pavia University dataset. Besides, we can observe that the proposed MambaHSI based on SSM outperforms all ML-based, GCN-based, CNN-based, and Transformer-based methods, and achieves SOTA on all datasets. This mainly benefits from Mamba's ability to model long-range dependencies with linear computational complexity, which enables MambaHSI to model more detailed long-range dependencies at the pixel level rather than at the patch level. The more detailed pixel-level long-range dependencies simultaneously provide detailed information (pixel level information) and global conception (long-range dependencies) to boost the discriminability of features for classification. Notably, to prove the potential of the SSM model in HSI classification, we only adopted the basic mamba layer as the basic unit, but did not adopt the newer and stronger mamba visual variant [19] designed for visual images. Even if only the basic mamba layer is used as the basic unit, the performance of the MambaHSI model has exceeded the existing methods based on ML, GCN, CNN, and Transformer. This demonstrates the great potential of SSM to be next-generation backbone for HSI model.

From the qualitative results shown in Figs. 7–10, we can observe that: in comparison with other methods, the proposed MambaHSI has less misclassification, producing a high-quality classification map with smooth objects and well-maintained boundaries. Take the HongHu dataset shown in Fig. 10 as example, SVM, RF, Spectralformer, GSC-ViT, and CLOLN contain a lot of small area noise. Meanwhile, DMSGer, GiGCN, and FullyContNet have large misclassification areas and fuzzy boundaries. In comparison with these methods, the classification map of the proposed method has better object integrity and clearer boundaries.

In summary, when compared with ML-based, GCN-based, CNN-based, and Transformer-based methods, the proposed MambaHSI based on SSM can achieve the best classification performance in accuracy and visualization. This reveals the great potential of SSM to be the next-generation backbone for HSI models.

C. Ablation Study

We conduct an ablation study to analyze the effect of each component, including SpaMB, SpeMB, and SSFM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

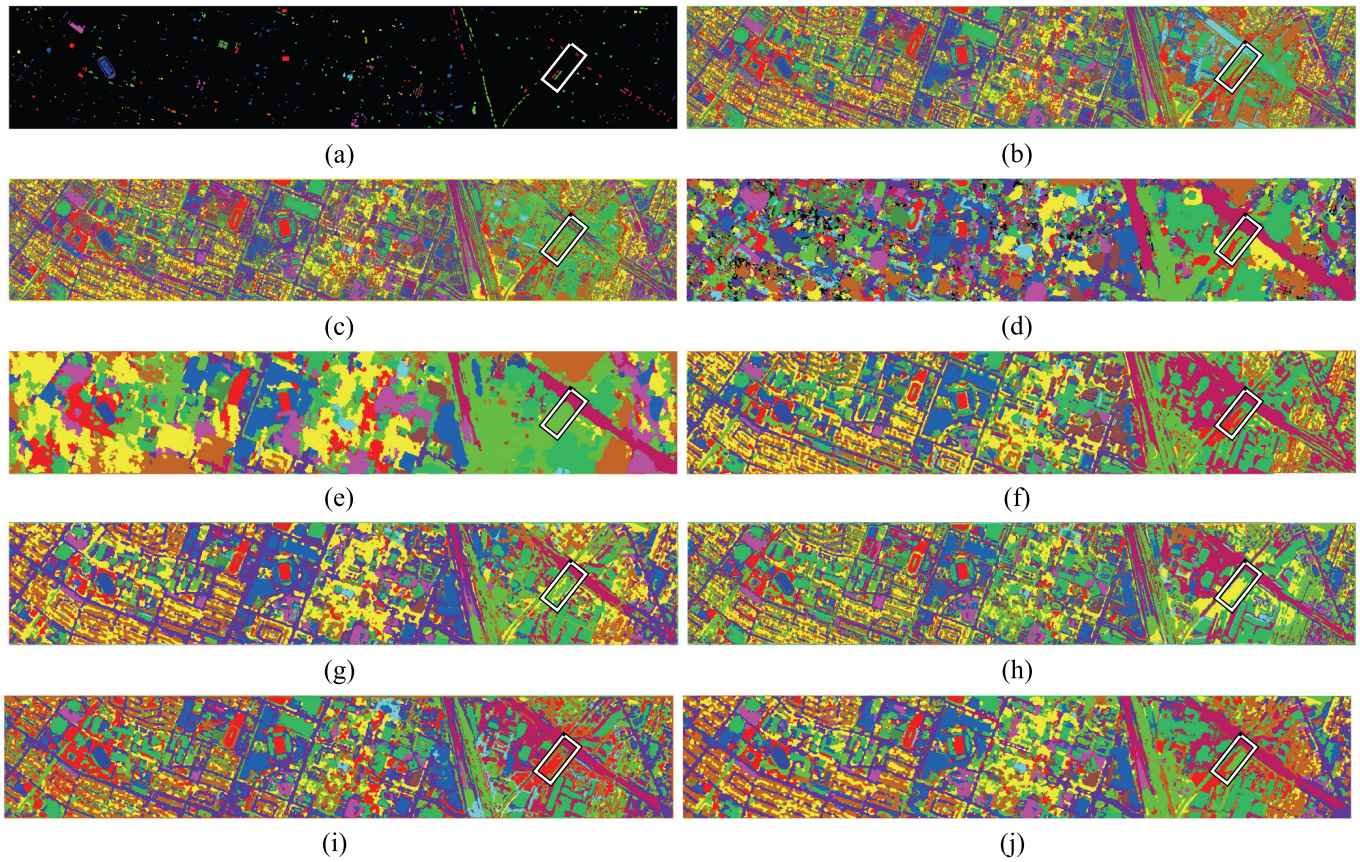


Fig. 8. Qualitative visualization of the classification map for Houston dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) DMSGer. (e) GiGCN. (f) FullyContNet. (g) CLOLN. (h) Spectralformer. (i) GSC-ViT. (j) Proposed MambaHSI. The meaning of colors refers to Fig. 4(c).

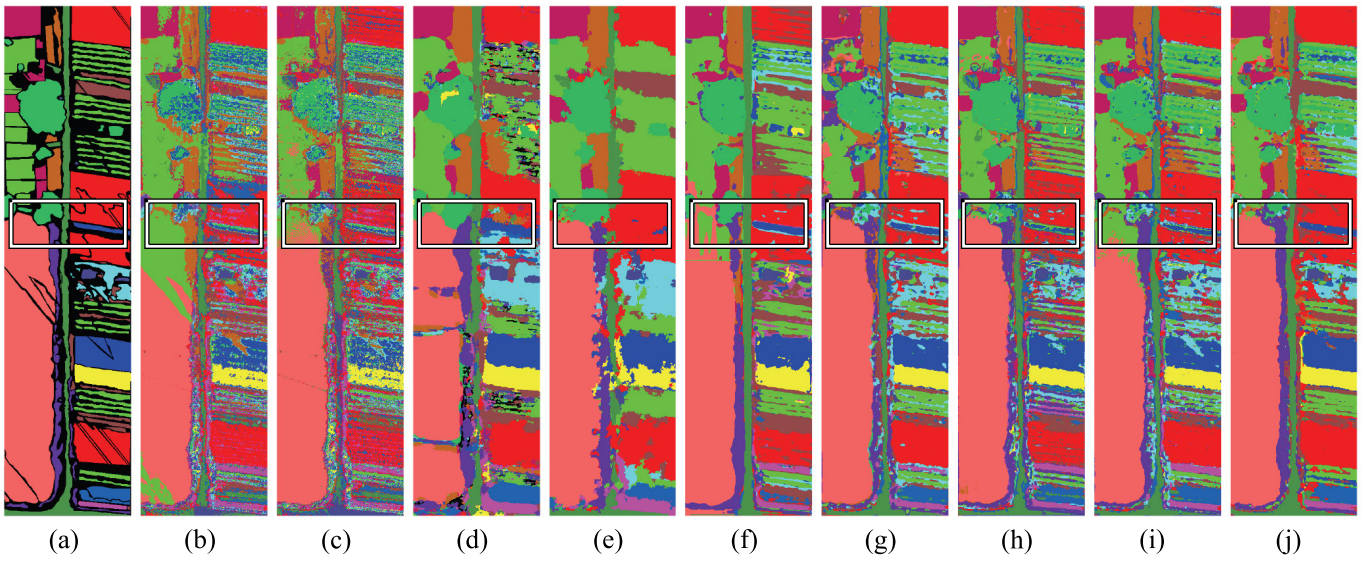


Fig. 9. Qualitative visualization of the classification map for HanChuan dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) DMSGer. (e) GiGCN. (f) FullyContNet. (g) CLOLN. (h) Spectralformer. (i) GSC-ViT. (j) Proposed MambaHSI. The meaning of colors refers to Fig. 5(c).

As shown in Table V, we can observe that: 1) the performance of spatial features exceeds that of spectral features, which indicates that spatial features are more discerning than spectral features; 2) directly summing spatial and spectral information does not always improve model performance, such as the performance degradation in Pavia University and HongHu datasets; and 3) after integrating spatial and spectral information by the proposed SSFM, the model performance is significantly enhanced. Taking HongHu dataset as example, the performance adding SSFM gains 5.47% improvement in OA over spatial information (SpaMB). This demonstrates the effectiveness of the proposed SSFM, and reveals that spatial features and spectral features are complementary.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

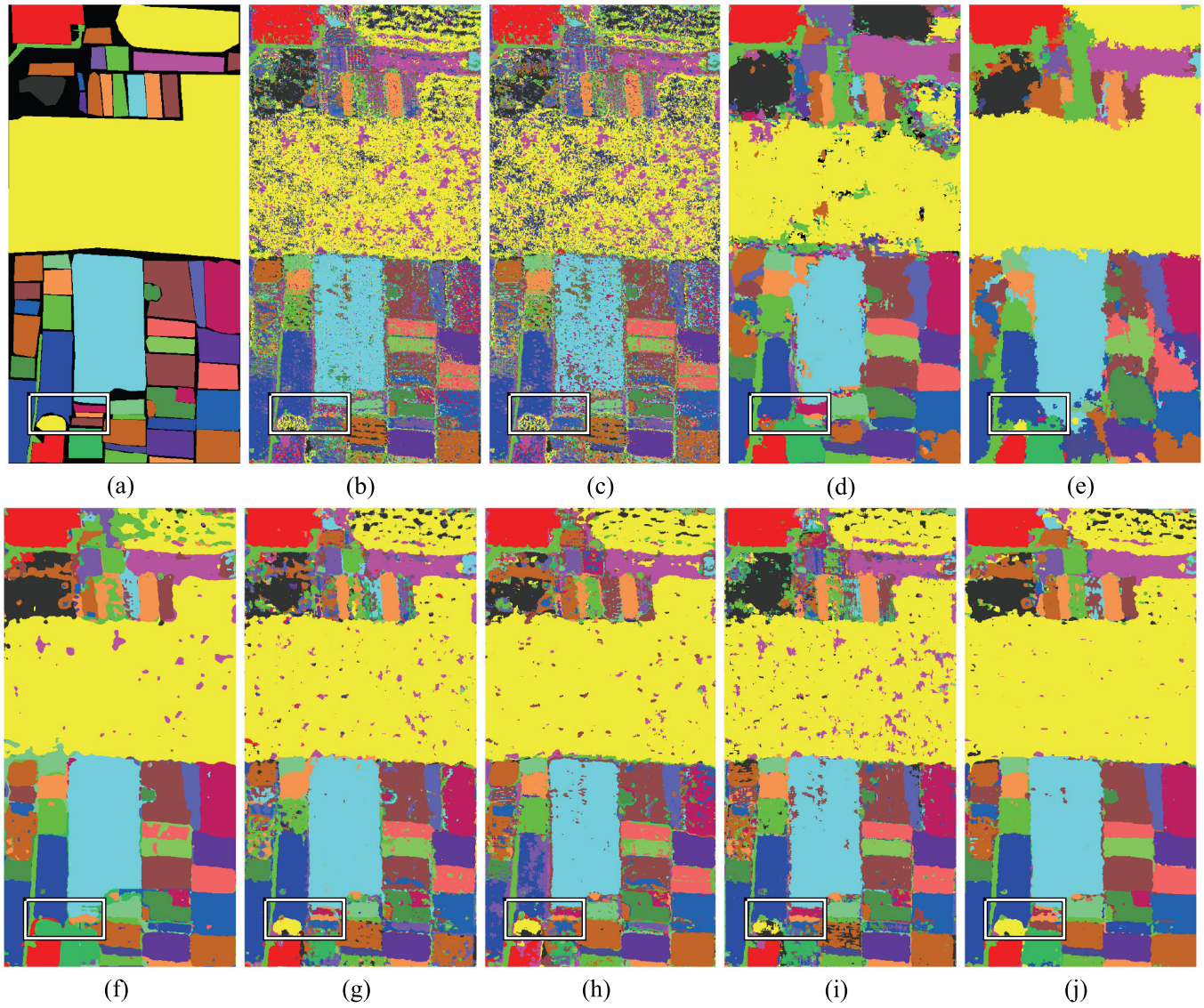


Fig. 10. Qualitative visualization of the classification map for HongHu dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) DMSGer. (e) GiGCN. (f) FullyContNet. (g) CLOLN. (h) Spectralformer. (i) GSC-ViT. (j) Proposed MambaHSI. The meaning of colors refers to Fig. 6(c).

TABLE V

ABLATION STUDY OF KEY COMPONENTS OF OUR METHOD. SPAMB, SPEMB, SUM, AND SSFM DENOTE THE SPAMB, SPECTRAL MAMBA BLOCK, SUMMATION OF SPATIAL AND SPECTRAL INFORMATION, AND SSFM, RESPECTIVELY

SpaMB	SpeMB	Sum	SSFM	PaviaU	Houston	HanChuan	HongHu
✓				95.63±0.95	92.56±1.31	87.84±3.85	89.11±1.88
	✓			78.34±4.00	81.33±2.55	73.19±2.79	66.97±4.41
✓	✓	✓		95.33±1.22	92.71±1.58	89.29±1.31	87.83±2.82
✓	✓		✓	95.74±0.90	94.46±0.83	90.21±1.67	94.58±1.01

D. Complexity Analysis

We compared the runtime, parameters, and floating point operations (FLOPs) of our method with several SOTA methods on the Pavia University dataset. From the results in Table VI, we observed that patch-level methods exhibit significantly higher computational complexity for full image processing compared to image-level methods. This is primarily due to the substantial redundant computations occurring between adjacent patches [3]. As shown in Table VII, these redundant computations also result in patch-level methods being significantly slower than image-level methods when testing on full images. Additionally, our method achieved the fastest testing and training times among image-level methods. Furthermore, we quantitatively compared the computational complexity of self-attention and Mamba. Specifically, we generated images of varying sizes and inputted them into an encoder block (EB) composed of a SpaMB, a SpeMB, and a SSFM to analyze the impact of image sequence length on Mamba's computational complexity. Subsequently, we replaced the Mamba layers in SpaMB and SpeMB with self-attention layers to further analyze the impact of image sequence length on the computational complexity of self-attention layers. As shown in Table VIII, we can observe that when the model's width and height are each doubled, resulting in a fourfold increase in image sequence length, the computational complexity of the encoder block using the Mamba layer also increases by approximately four times. This quantitatively demonstrates the linear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

COMPARISON OF COMPUTATION COST ON THE PAVIA UNIVERSITY DATASET. $\text{FLOPs}_{\text{image}}$ DENOTES THE FLOPS TO TEST A WHOLE HSI. *M* AND *G* SIGNIFY MEGA AND GIGA, RESPECTIVELY. † AND ‡ DENOTE THE PATCH-LEVEL METHOD AND IMAGE-LEVEL APPROACH, RESPECTIVELY

Metrics	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM† TGRS2004	RF† TGRS2005	DMSGer‡ TNNLS2022	GiGCN‡ TNNLS2022	FullyContNet‡ TGRS2022	CLOLN† TGRS2024	Spectralformer† TGRS2021	GSC-ViT† TGRS2024	MambaHSI‡ Ours
Paramters (M)	-	-	0.090	0.223	1.200	0.005	0.177	0.153	0.412
$\text{FLOPs}_{\text{image}}$ (G)	-	-	7.66	11.01	145.62	124.44	9086.19	2026.298	39.18

TABLE VII

COMPARISON OF RUNNING TIME ON THE PAVIA UNIVERSITY DATASET. T_{TR} DENOTES THE TRAINING TIME, WHILE T_{TE} DENOTES THE TIME TO TEST A WHOLE HSI. *s* DENOTES SECONDS. † AND ‡ DENOTE THE PATCH-LEVEL METHOD AND IMAGE-LEVEL APPROACH, RESPECTIVELY

Metrics	ML-based		GCN-based		CNN-based		Transformer-based		SSM-based
	SVM† TGRS2004	RF† TGRS2005	DMSGer‡ TNNLS2022	GiGCN‡ TNNLS2022	FullyContNet‡ TGRS2022	CLOLN† TGRS2024	Spectralformer† TGRS2021	GSC-ViT† TGRS2024	MambaHSI‡ Ours
T_{tr} (s)	0.01	0.29	576.88	556.31	612.15	38.43	94.19	62.21	274.54
T_{te} (s)	3.52	1.96	0.09	3.78	0.09	46.46	62.43	49.44	0.03

TABLE VIII

QUANTITATIVE COMPARISON OF COMPUTATIONAL COMPLEXITY BETWEEN MAMBA AND TRANSFORMER. THE GFLOPS ARE REPORTED

Model	Mamba	Self-Attention	25×25	50×50	100×100	200×200
EB	✓		0.08	0.32	1.28	5.13
EB		✓	0.28	3.54	52.87	830.66

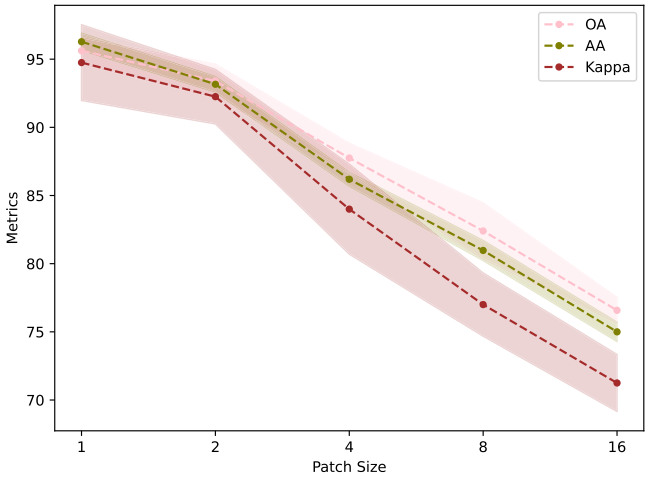


Fig. 11. Effect of replacing pixel-wise spatial features with patch features of different sizes on the Pavia University dataset.

complexity of Mamba. Additionally, when the image size increased from 100×100 to 200×200, the sequence length increased fourfold. Using an encoder block with self-attention layers increased the computational complexity by approximately sixteen fold. This quantitatively demonstrates the quadratic complexity of self-attention layers.

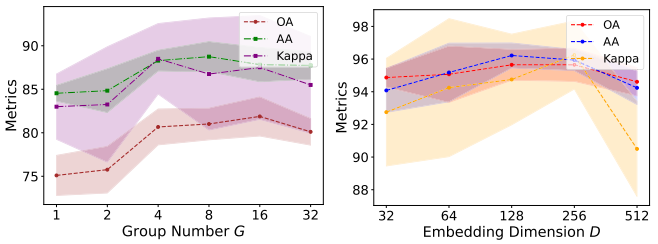
E. Effect of Pixel-Wise Spatial Features

To analyze the effect of pixel-wise spatial features, we replaced the pixel-wise features with the patch features of different patch sizes to train MambaHSI and compared them with pixel-wise features. From the results in Fig. 11, we can observe that as the patch size increases, the model performance degrades significantly. This demonstrates the importance of pixel-wise features.

TABLE IX

EFFECT OF SSI ON THE PAVIA UNIVERSITY DATASET

Models	OA	AA	Kappa
w/o SSI	75.10±2.31	84.54±0.92	83.00±3.76
w SSI	80.67±2.07	88.29±1.21	88.50±4.06



(a) (b)

Fig. 12. Hyperparameter analysis on the Pavia University dataset. (a) Effect of group number G. (b) Effect of embedding dimension D. The figure reports the mean and standard deviation of the results from five runs.

F. Effect of Spectral Sequence Information

To evaluate the effect of the spectral sequence information, the group G of spectral mamba is set to 1. Because there is only one group ($G = 1$), no sequence order relationship between spectra is introduced. As shown in Table IX compared to the spectral mamba with introduced sequential spectral information, the Spectral Mamba without spectral sequence information shows performance degradation of 5.57%, 3.75%, and 5.50% in terms of OA, AA, and Kappa, respectively. This indicates that introducing the spectral sequence information is beneficial for HSI classification.

G. Hyperparameter Analysis

We conducted the hyperparameter analysis on the Pavia University dataset. From the results in Fig. 12(a), we can observe that the performance of Spectral Mamba is the worst when the number of groups for Spectral Mamba is set to 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

This proves the effectiveness of introducing spectral continuity by sequentially inputting multiple groups of spectra into Mamba. Additionally, when the number of spectrum groups is greater than four, the performance becomes relatively stable. From the results in Fig. 12(b), we find that as the number of hidden dimensions increases, model performance initially improves and then declines. Initially, the increase in dimensions enhances the model's learning capacity, resulting in improved performance. However, as the dimensions continue to increase, the sample size of the hyperspectral dataset becomes insufficient to support the model's learning capacity, leading to overfitting and a subsequent decline in performance.

VI. Conclusion

In this article, we present MambaHSI, the first SSM-based image-level HSI classification method, which has strong capability of modeling long-range dependencies and integrating spectral–spatial information. MambaHSI introduces Mamba as a basic unit to model the long-range interactions of the whole image, which enables the model to capture the long-range dependencies of the whole HSI image while maintaining the linear computational complexity. The proposed SpaMB, spectral Mamba block, and SSFM enable the model to mine discriminative spatial and spectral features and then adaptively fuse them for HSI classification. From the extensive experimental results on multiple datasets, we mainly conclude that: 1) SSM has great potential to be the next-generation backbone for HSI classification benefiting from the strong capability of modeling long-range dependencies while maintaining linear computational complexity and 2) integrating spatial and spectral information is vital for HSI classification. In the future, we intend to extend the proposed idea into more HSI tasks, such as weakly supervised HSI classification and HSI clustering.

References

[1] Q. Zhu et al., "A spectral-spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[4] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[5] S. Pande and B. Banerjee, "HyperLoopNet: Hyperspectral image classification using multiscale self-looping convolutional networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 422–438, Jan. 2022.

[6] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[7] Z. Wang, B. Du, L. Zhang, L. Zhang, and X. Jia, "A novel semisupervised active-learning algorithm for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3071–3083, Jun. 2017.

[8] C. Fu, B. Du, and L. Zhang, "Do we need learnable classifiers? A hyperspectral image classification algorithm based on attention-enhanced ResBlock-in-ResBlock and ETF classifier," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505013.

[9] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[10] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[11] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[12] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[13] Y. Yang et al., "Semi-supervised multiscale dynamic graph convolution network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6806–6820, May 2024.

[14] C. Fu, B. Du, and L. Zhang, "ReSC-Net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[15] C. Li, B. Rasti, X. Tang, P. Duan, J. Li, and Y. Peng, "Channel-layer-oriented lightweight spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504214.

[16] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[17] J. Ruan and S. Xiang, "VM-UNet: Vision mamba UNet for medical image segmentation," 2024, *arXiv:2402.02491*.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[19] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[20] A. Gu, *Modeling Sequences With Structured State Spaces*. Stanford, CA, USA: Stanford Univ. Press, 2023.

[21] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.

[22] A. Gu, K. Goel, and C. Re, "Efficiently modeling long sequences with structured state spaces," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–27.

[23] K. Goel, A. Gu, C. Donahue, and C. Ré, "It's raw! Audio generation with state-space models," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 7616–7633.

[24] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[25] J. Ma, F. Li, and B. Wang, "U-mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, *arXiv:2401.04722*.

[26] R. Xu, S. Yang, Y. Wang, Y. Cai, B. Du, and H. Chen, "Visual mamba: A survey and new outlooks," 2024, *arXiv:2404.18861*.

[27] Y. Oshima, S. Taniguchi, M. Suzuki, and Y. Matsuo, "SSM meets video diffusion models: Efficient video generation with structured state spaces," 2024, *arXiv:2403.07711*.

[28] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," 2024, *arXiv:2403.19654*.

[29] D. Liang et al., "PointMamba: A simple state space model for point cloud analysis," 2024, *arXiv:2402.10739*.

[30] B. Tu, J. Wang, X. Kang, G. Zhang, X. Ou, and L. Guo, "KNN-based representation of superpixels for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4032–4047, Nov. 2018.

[31] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.

[32] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.

[33] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[34] L. Fang, S. Li, X. Kang, and J. A. Benediktsson, "Spectral–spatial classification of hyperspectral images with a superpixel-based discriminative sparse model," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4186–4201, Aug. 2015.

[35] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[36] Y. Gu, C. Wang, D. You, Y. Zhang, S. Wang, and Y. Zhang, "Representative multiple kernel learning for classification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 7, pp. 2852–2865, Jul. 2012.

[37] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.

[38] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2018.

[39] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[40] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.

[41] M. Ahmad et al., "Hyperspectral image classification-traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[42] J. Yang, B. Du, and L. Zhang, "From center to surrounding: An interactive learning framework for hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 145–166, Mar. 2023.

[43] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Nov. 2015.

[44] S. Li, X. Zhu, and J. Bao, "Hierarchical multi-scale convolutional neural networks for hyperspectral image classification," *Sensors*, vol. 19, no. 7, p. 1714, Apr. 2019.

[45] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] D. Wang, B. Du, and L. Zhang, "Fully contextual network for hyperspectral scene parsing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501316.

[47] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[48] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.

[49] Z. Li, Z. Xue, Q. Xu, L. Zhang, T. Zhu, and M. Zhang, "SPFormer: Self-pooling transformer for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502019.

[50] J. T. Smith, A. Warrington, and S. Linderman, "Simplified state space layers for sequence modeling," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–35.

[51] D. Y. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Re, "Hungry hungry hippos: Towards language modeling with state space models," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–25.

[52] H. Mehta, A. Gupta, A. Cutkosky, and B. Neyshabur, "Long range language modeling via gated state spaces," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–20.

[53] E. Nguyen et al., "S4ND: Modeling images and videos as multidimensional signals with state spaces," in *Proc. Adv. Neural Inf. Process. Syst.*, A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho, Eds., 2022, pp. 1–16.

[54] M. M. Islam, M. Hasan, K. S. Athrey, T. Braskich, and G. Bertasius, "Efficient movie scene detection using state-space transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18749–18758.

[55] J. Wang et al., "Selective structured state-spaces for long-form video understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6387–6397.

[56] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[57] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," 2024, *arXiv:2405.12487*.

[58] L. Huang, Y. Chen, and X. He, "Spectral–spatial mamba for hyperspectral image classification," 2024, *arXiv:2404.18401*.

[59] J. Sheng, J. Zhou, J. Wang, P. Ye, and J. Fan, "DualMamba: A lightweight spectral–spatial mamba-convolution network for hyperspectral image classification," 2024, *arXiv:2406.07050*.

[60] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[61] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6029–6043, Jul. 2021.

[62] W. Qi, C. Huang, Y. Wang, X. Zhang, W. Sun, and L. Zhang, "Global–local 3-D convolutional transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510820.

[63] G. Sun et al., "Large kernel spectral and spatial attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519915.

[64] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.



Yapeng Li (Graduate Student Member, IEEE) received the B.S. degree from Taiyuan University of Technology, Taiyuan, China, in 2017, and the M.E. degree from the School of Geodesy and Geomatics, Wuhan University, Wuhan, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Computer Science, Wuhan University.

His research interests include remote sensing, computer vision, and multimodal learning.



Yong Luo (Member, IEEE) received the B.E. degree in computer science from Northwestern Polytechnical University, Xi'an, China, in 2009, and the D.Sc. degree from the School of Electronics Engineering and Computer Science, Peking University, Beijing, China, in 2014.

He is currently a Professor with the School of Computer Science, Wuhan University, Wuhan, China. He has authored or co-authored more than 60 papers in top journals and prestigious conferences including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), and IEEE TRANSACTIONS ON MULTIMEDIA (TMM), ICCV, WWW, IJCAI, and AAAI. His research interests include machine learning and data mining with applications to visual information understanding and analysis.

Dr. Luo was a recipient of the IEEE Globecom 2016 Best Paper Award and was nominated for the IJCAI 2017 Distinguished Best Paper Award. He was also a co-recipient of the IEEE ICME 2019 and IEEE VCIP 2019 Best Paper Awards. He is serving on the Editorial Board for IEEE TRANSACTIONS ON MULTIMEDIA.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.



Lefei Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively.

He was a Big Data Institute Visitor with the Department of Statistical Science, University College London, London, U.K., and a Hong Kong Scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China. He is currently a Professor with the School of Computer Science, Wuhan University, and also with Hubei Luojia Laboratory, Wuhan. His research interests include pattern recognition, image processing, and remote sensing.

Dr. Zhang serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and an Associate Editor for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Zengmao Wang (Member, IEEE) received the M.S. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, in 2015, and the Ph.D. degree from the School of Computer Science, Wuhan University, in 2019.

He is funded by the National Post-Doctoral Program for Innovative Talents and has been a Postdoctoral Researcher with Wuhan University since 2019. He has authored or co-authored more than 30 research papers published in IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEM (TNNLS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), ECCV, and IJCAI. His research interests include image processing and machine learning.



Bo Du (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2010.

He is currently a Professor with the School of Computer Science and the Institute of Artificial Intelligence, Wuhan University, where he is also the Director of the National Engineering Research Center for Multimedia and Software. He has published more than 80 research articles in IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). Thirteen of them are Essential Science Indicators (ESI) hot articles or highly cited articles. His research interests include pattern recognition, hyperspectral image processing, and signal processing.

Dr. Du serves as a Senior Program Committee (PC) Member of the International Joint Conferences on Artificial Intelligence (IJCAI) and the Association for the Advancement of Artificial Intelligence (AAAI). He received the Highly Cited Researcher by the Web of Science Group in 2019 and 2020, the IEEE Geoscience and Remote Sensing Society (GRSS) 2020 Transactions Prize Paper Award, the IJCAI Distinguished Paper Prize, the IEEE Data Fusion Contest Champion, and the IEEE Workshop on Hyperspectral Image and Signal Processing Best Paper Award in 2018. He has served as the Area Chair for the International Conference on Pattern Recognition (ICPR). He serves as an Associate Editor for *Neural Networks*, *Pattern Recognition*, and *Neurocomputing*. He also serves as a reviewer for 20 Science Citation Index (SCI) magazines, including IEEE TPAMI, IEEE TCYB, IEEE TGRS, IEEE TIP, IEEE JSTARS, and IEEE GRSL.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:38:03 UTC from IEEE Xplore. Restrictions apply.